Name of the Student	K. Lokesh Chandra
Reg. No	192425358
GitHub Link	

SIMATS ENGINEERING

CO3 - ASSESSMENT TOOL 3

Comparative Analysis

COURSE NAME: **DEEP LEARNING** **COURSE CODE:** **MLA04**
SLOT :

COURSE OUTCOME COVERED

CO3: Students will be able to understand, analyze and apply CNN concepts including layers, filters, parameter sharing, regularization, AlexNet and ResNet for real-world image processing applications. (BTL-6)

CO Weightage: 25% **Duration: 90 minutes**

Assessment Tool Weightage: 40% **Marks: 25**

Code of Conduct:

I ____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **Name of the student:**

Criteria	Selection of Studies/Parameters (2)	Comparison & Differentiation (2.5)	Critical Evaluation (2.5)	Synthesis & Insight Generation (2)	Clarity & Presentation *(1)*	Total (10)
Weightage	*20 %*	*25%*	*25%*	*20 %*	*10 %*	100 %
Q1						
Q2						
Q3						
Q4						
Q5						
Total						

Signature of the Faculty: ***Total Marks :***

QUESTIONS: 05 **Total Marks:25**

Q1. CNN vs Fully Connected Neural Network

A medical imaging organization wants to classify X-ray images using deep learning. Compare a **CNN with a traditional fully connected neural network** in terms of architectural design, motivation, spatial feature extraction, number of parameters, computational complexity, and ability to handle image data. Analyze the roles of convolutional layers, filters, and pooling layers, and justify why CNNs are more suitable for image classification applications.

Q2. Different CNN Layers and Filters

A manufacturing company is developing a CNN-based system to detect defects in industrial products. Compare the functions of **convolutional layers, pooling layers, activation layers, and fully connected layers** in terms of feature extraction, spatial information, computational requirements, and classification. Further compare **early-layer filters and deeper-layer filters** based on the type of features they learn, and justify how these layers collectively improve defect detection.

Q3. Parameter Sharing vs Independent Weights

A smart surveillance system needs to process thousands of high-resolution images efficiently. Compare **parameter sharing in CNNs with independent weights in fully connected networks** in terms of the number of parameters, memory requirements, computational complexity, feature detection, and translation-related invariance. Analyze how the reuse of convolutional filter weights improves learning efficiency and justify its importance for large-scale image-processing applications.

Q4. AlexNet vs ResNet

A computer vision company is developing an image classification system and is considering **AlexNet and ResNet** as possible architectures. Compare the two architectures in terms of network depth, layer organization, convolutional filters, parameter requirements, feature extraction capability, training difficulty, vanishing-gradient problem, computational cost, and classification performance. Recommend the more suitable architecture for a complex image recognition application and justify your choice.

Q5. Regularized CNN vs Non-Regularized CNN

An agricultural organization is developing a CNN to classify plant diseases from leaf images. The initial model achieves very high training accuracy but performs poorly on unseen images. Compare a **CNN without regularization and a CNN using dropout, data augmentation, and batch normalization** in terms of overfitting, generalization, training stability, computational requirements, and model performance. Analyze which regularization techniques should be applied and justify the most appropriate approach for reliable real-world disease classification.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Selection of Studies/Parameters	20%	Selects highly relevant and diverse studies with well-defined comparison parameters	Selects relevant studies with minor gaps in parameter definition	Limited or partially relevant selection	Poor or inappropriate selection of studies
Comparison & Differentiation	25%	Clearly compares multiple aspects (methods, results, limitations) with strong differentiation	Good comparison with minor gaps	Basic comparison with limited depth	No clear comparison; mostly descriptive
Critical Evaluation	25%	Critically evaluates strengths, weaknesses, and implications with strong justification	Provides evaluation with some justification	Limited evaluation; lacks depth	No critical evaluation provided
Synthesis & Insight Generation	20%	Generates meaningful insights and conclusions from comparisons	Some insights derived from comparison	Limited or unclear insights	No insights generated
Clarity	10%	Information is clearly presented (tables/figures) with logical structure	Generally clear with minor issues	Presentation lacks clarity or structure	Poor presentation and difficult to understand

1. CNN vs Fully Connected Neural Network

A medical imaging organization wants to classify X-ray images. CNNs are specially designed for image data, while traditional fully connected neural networks treat the input mainly as a collection of individual values.

Comparison

Parameter	**CNN**	**Fully Connected Neural Network**
Architecture	Uses convolution and pooling layers	Uses fully connected layers
Spatial information	Preserves spatial relationships	Spatial relationships are mostly lost after flattening
Feature extraction	Automatically extracts image features	Requires more direct learning from flattened input
Parameters	Fewer because of parameter sharing	Very large number of parameters
Computation	More efficient for images	High for large images
Image handling	Highly suitable	Less suitable
Translation detection	Can detect features at different positions	Less effective
Main application	Image and video processing	General classification/regression

Role of CNN Layers

Convolutional layer: Extracts important image features such as edges, textures and shapes.

Filters: Small learnable matrices that scan the image and detect specific patterns.

Pooling layer: Reduces the spatial size of feature maps and decreases computational requirements.

Analysis

Suppose an X-ray image has thousands of pixels. A fully connected network would create a large number of connections between pixels and neurons. This increases parameters, memory and computation.

CNNs use local connections and shared filters. Therefore, the same filter can detect a particular feature in different areas of the X-ray.

Conclusion

CNN is more suitable for X-ray classification. It preserves spatial information, uses fewer parameters and automatically learns useful image features. A fully connected network can perform classification, but it becomes inefficient when the image size is large.

Insight: CNN's ability to combine **local feature extraction and parameter sharing** makes it much more suitable for image classification.

2. Different CNN Layers and Filters

The manufacturing company wants to detect defects in industrial products. Different CNN layers perform different functions and work together to identify defects.

Comparison of CNN Layers

Layer	Main Function	Feature Learning	Computational Requirement
Convolution	Extracts features	Edges, textures, shapes	Moderate
Activation	Adds non-linearity	Complex patterns	Low
Pooling	Reduces feature-map size	Keeps important information	Low
Fully Connected	Performs final classification	Combines learned features	Higher

Convolution Layer

The convolution layer applies filters to different parts of the product image. It can detect:

- Edges
- Lines
- Textures
- Shapes
- Surface patterns

This is useful for detecting scratches, cracks and other defects.

Activation Layer

The ReLU activation function introduces non-linearity. It allows the CNN to learn complex relationships instead of only simple linear patterns.

Pooling Layer

Pooling reduces the size of feature maps.

Its advantages are:

- Reduces computation.

- Reduces memory requirements.
- Retains important features.
- Makes the model less sensitive to small changes.

Fully Connected Layer

The fully connected layer receives the extracted features and performs the final classification, such as:

Normal product → No defect

Defective product → Defect detected

Early vs Deep Filters

Early-layer filters usually learn simple features:

- Edges
- Lines
- Corners
- Basic textures

Deeper-layer filters learn complex features:

- Shapes
- Object parts
- Defect patterns
- Complete abnormal structures

Critical Evaluation

If only early layers are used, the model may detect simple edges but may not understand the complete defect. If deeper layers are used, the model can understand more complex patterns but requires more computation.

Therefore, a combination of convolution, activation, pooling and classification layers provides a better solution.

Conclusion

The layers work progressively:

Image → Edges → Textures → Shapes → Defect patterns → Classification

Thus, combining different CNN layers improves the accuracy and reliability of industrial defect detection.

3. Parameter Sharing vs Independent Weights

The surveillance system needs to process thousands of high-resolution images. Therefore, computational efficiency and memory usage are important.

Comparison

Parameter	**CNN Parameter Sharing**	**Fully Connected Independent Weights**
Weights	Same filter reused	Different weights for connections
Number of parameters	Low	Very high
Memory	Lower	Higher
Computation	Lower	Higher
Feature detection	Efficient	Less efficient for images
Position handling	Detects features at different locations	Less naturally suited
Large images	Highly suitable	Expensive

Parameter Sharing

In CNNs, one filter is used repeatedly across different locations.

For example, a **3 × 3 filter** has only:

9 weights + 1 bias

The same filter can scan the complete image.

This means the CNN does not need a completely different set of weights for every image location.

Independent Weights

In a fully connected network, each connection has its own weight.

For a high-resolution image, this can produce millions of parameters.

This results in:

- High memory usage.
- More computation.
- Longer training.
- Greater possibility of overfitting.

Critical Evaluation

Parameter sharing does not mean that the CNN uses only a few features. Instead, the same learned filter is applied across different positions.

For example, a filter that learns to detect an edge can detect that edge whether it appears on the left, right, top or bottom of a surveillance image.

This provides useful translation-related feature detection.

Conclusion

Parameter sharing is extremely important for large-scale image processing because it reduces parameters and computational requirements while allowing useful features to be detected at different positions.

Insight: Parameter sharing provides a good balance between **feature detection and computational efficiency**, making CNNs practical for thousands of high-resolution surveillance images.

4. AlexNet vs ResNet

A computer vision company is considering AlexNet and ResNet for an image classification system. The assessment requires comparison of depth, layer organization, filters, parameters, feature extraction, training, gradients, computation and performance.

Comparison

Parameter	**AlexNet**	**ResNet**
Network depth	Shallower	Much deeper
Architecture	Traditional CNN	CNN with residual connections
Filters	Convolution filters	Convolution filters at many layers
Feature extraction	Good	Very strong
Training	Relatively easier	More complex but deep training is easier
Vanishing gradient	More problematic in deeper networks	Reduced using skip connections
Parameters	Large for its time	Deep models can have many parameters, but architecture is more effective
Computation	Lower	Generally higher
Classification	Good	Generally better for complex tasks

AlexNet

AlexNet uses convolutional layers followed by other layers for classification. It was an important development in deep learning and provides good image classification performance.

Advantages:

- Simpler architecture.
- Easier to understand.
- Lower computational requirement than many deeper models.

Limitations:

- Limited depth compared with ResNet.
- Less capable of learning very complex hierarchical features.
- Deepening the network can create gradient problems.

ResNet

ResNet introduces **residual/skip connections**. These connections allow information to pass directly across layers.

Advantages:

- Supports much deeper networks.
- Better feature extraction.
- Reduces the vanishing-gradient problem.
- Suitable for complex image recognition.

Limitations:

- Higher computational requirement.
- More complex architecture.
- May require hardware optimization for real-time applications.

Critical Evaluation

For simple image classification, AlexNet may be sufficient and computationally easier.

For a complex recognition problem containing many different objects and visual patterns, ResNet provides stronger feature learning.

Recommendation: ResNet

ResNet is the better choice for a complex image recognition application.

Its deeper architecture and residual connections allow it to learn more complex features and reduce training problems associated with deep networks.

However, computational cost should be considered. If real-time performance is required, an optimized ResNet version should be selected.

5. Regularized CNN vs Non-Regularized CNN

The agricultural organization has a CNN that gives **very high training accuracy but poor performance on unseen images**. This clearly indicates an **overfitting problem**. The assessment asks for comparison of regularized and non-regularized CNNs and selection of suitable techniques.

Comparison

Parameter	**Non-Regularized CNN**	**Regularized CNN**
Overfitting	High	Lower
Generalization	Poor	Better
Training accuracy	May be very high	Usually more realistic
Validation performance	Often poor	Usually better
Training stability	May be unstable	More stable with suitable methods
Computation	Lower	Slightly higher
Real-world performance	Poorer	Better

1. Dropout

Dropout randomly disables some neurons during training.

Advantages:

- Reduces dependency on individual neurons.
- Reduces overfitting.
- Improves generalization.

For plant disease classification, dropout can prevent the model from memorizing specific leaf images.

2. Data Augmentation

Data augmentation creates different versions of existing leaf images.

Examples include:

- Rotation
- Scaling
- Cropping
- Brightness changes
- Small shifts

This is very useful because leaves can appear at different angles, sizes and lighting conditions.

Advantage

The CNN sees more variations during training and becomes better at recognizing diseases in unseen images.

3. Batch Normalization

Batch normalization normalizes the activations during training.

It can:

- Stabilize training.
- Improve convergence.
- Support better generalization.
- Allow more reliable learning.

Critical Evaluation

The original CNN has high training accuracy but poor unseen-image performance. Therefore, simply increasing training time is not a good solution because it may increase overfitting.

A better approach is to use **data augmentation and dropout**, with **batch normalization** to improve training stability.

Recommended Approach

Data Augmentation + Batch Normalization + Dropout

This combination provides a good balance:

- Data augmentation → More varied training data.
- Batch normalization → More stable training.
- Dropout → Less dependency on specific neurons.

Conclusion

The **regularized CNN is more suitable** for real-world plant disease classification because it can generalize better to new leaf images.

The main goal should not be only high training accuracy. The model should achieve **good validation and testing performance** because real agricultural images will be different from the training images.

Insight: A slightly lower training accuracy with much better unseen-image performance is preferable to extremely high training accuracy with poor generalization.